Ahmed Fathy

Software Engineer

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Education

- **Cairo University, Faculty of Engineering** — Giza, Egypt
  B.Sc. in Computer Engineering; GPA: 3.7 — *2022 – 2027*

Experience

- **Al Nada Scientific Office** | *Software Developer (Part-time)* — *Cairo, Egypt | June 2022 – 2025*
  - **ERP System**: Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
  - **Corporate Web Solution**: Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

Projects

- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* — *2025*
  - **Core Architecture**: Designed a 32-bit 5-stage pipelined processor in VHDL, with ISA of over 25 instructions.
  - **Hazard Resolution**: Implemented Hazard Detection and Data Forwarding units to resolve control and data dependencies.
  - **Verification**: Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.

- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* — *2025*
  - **Architecture**: Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
  - **Memory Hierarchy**: Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
  - **Physical Design**: Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* — *2023*
  - **Core Logic**: Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
  - **Datapath**: Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
  - **Validation**: Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- **Tactic (Robotics & AI)** | *Embedded Systems* — *2024*
  - **System Integration**: Developed browser-controlled robotics with AI vision and low-latency control
  - **Firmware**: Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
  - **Backend Processing**: Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- **Autonomous Wall-Following Robot** | *Embedded Systems & C/C++* — *2026*
  - **System Architecture**: Designed a 5-state FSM to manage autonomous navigation, turning, and data flow.
  - **Low-Level Firmware**: Programmed STM32 microcontroller at the register level to optimize hardware timer usage.
  - **Control & Integration**: Combined proportional control, ultrasonic/gyro sensors, and WiFi Module for tracking.
- **Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* — *2024*
  - **Pipeline**: Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
  - **Algorithms**: Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- **Machine Fault Listener System** | *Machine Learning & Audio Processing* — *2026*
  - **Pipeline & Architecture**: Created an end-to-end Machine Learning pipeline to classify industrial audio into six operational states.
  - **Optimization**: Optimized a deep learning model for accuracy and fast inference within a 2B-parameter limit.

Student Activities

- **Physics Helping District (PhD)** | *Team Leader* — *Cairo | June 2022 – 2024*
  - **Mentorship**: Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* — *Cairo | Oct 2024*
  - **Orrery App**: Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

Technical Skills

- **Digital Design & Verification**: Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design

- **Development Tools & Platforms**: ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages**: C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly